

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Flywheel capscrew socket, Part Number 2892191
- Crack Detection Kit, Part Number 3375432

Additional Service Items

- Hoist
- Two T-handles
- Lifting sling
- Soft-faced mallet
- Lubriplate™ 105 multi-purpose lubricant

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

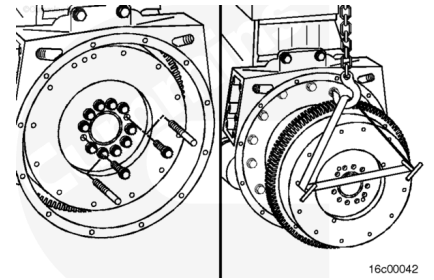
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries or air starter to prevent accidental engine starting. See equipment manufacturer service information.
- Remove the transmission, clutch, and all related components. See equipment manufacturer service information.

Remove

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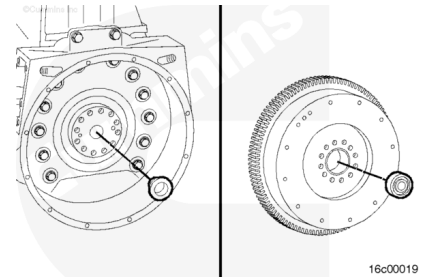
Use two M16 x 1.5 guide studs to prevent the flywheel from rotating. Use flywheel cap screw socket, Part Number 2892191, to remove two cap screws and install the guide studs.

Use a hoist, two t-handles, and a lifting sling. Install the tee-handles. Use flywheel cap screw socket, Part Number 2892191, to remove the remaining ten cap screws. Remove the flywheel.

Use a soft-faced mallet to tap the flywheel from the crankshaft, if necessary. Damage to the clutch surface can occur if a hard-faced hammer is used.

Remove the pilot bearing spacer from the rear of the crankshaft.

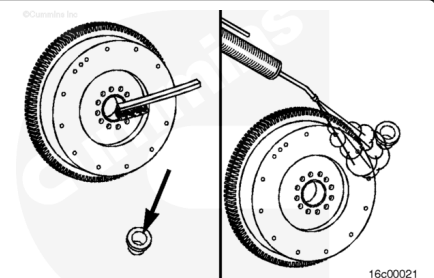
Use a mandrel and hammer to remove the pilot bearing from the flywheel.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a wire brush to clean the crankshaft pilot bore.

Use a wire brush to clean the flywheel pilot bore.

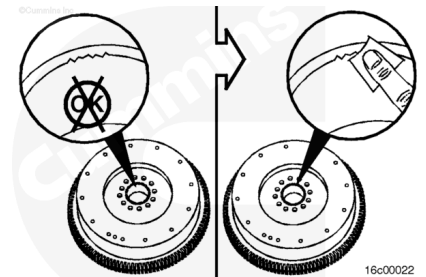
Clean the flywheel with steam or solvent.

Dry with compressed air.

Inspect the flywheel and pilot bores for nicks or burrs.

Use a fine crocus cloth to remove small nicks and burrs.

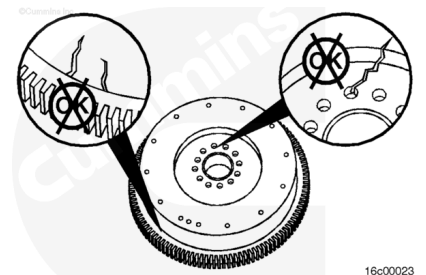
For information on flywheel resurfacing, see Service Bulletin, Minimum Flywheel Dimensions for Resurfacing and Reuse, Bulletin 6506608
(/qs3/pubsys2/xml/en/bulletin/6506608.html).



⚠ WARNING ⚠

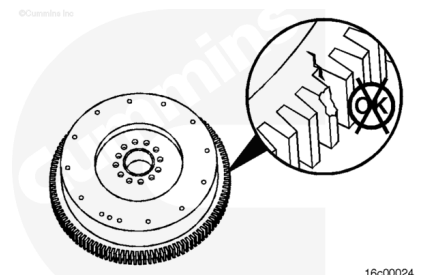
Do not use a cracked flywheel. A cracked flywheel can break and cause personal injury or property damage.

Use the crack detection kit, Part Number 3375432, to check for cracks in the flywheel. Follow the instructions provided with the kit.



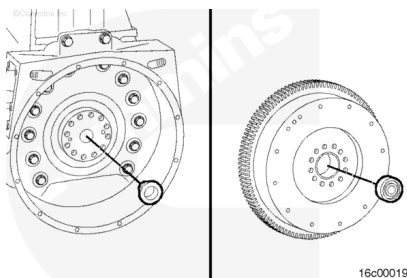
Inspect the flywheel ring gear teeth for cracks and chips.

If the ring gear teeth are cracked or broken, the ring gear **must** be replaced. Refer to Procedure 016-008 in Section 16.
(/qs3/pubsys2/xml/en/procedures/377/377-016-008.html)



Inspect the pilot bearing spacer for cracks or other damage.

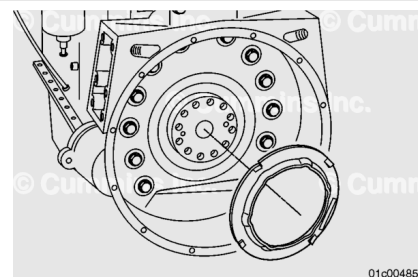
Replace the pilot bearing spacer, if necessary.



Install

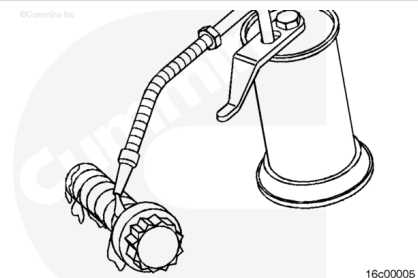
Install the pilot bearing spacer into the rear of the crankshaft with the spacer lip oriented outward.

Note : A small amount of clean Lubriplate™ 105 multi-purpose lubricant, or its equivalent, can be used to help keep the pilot bearing spacer in position until the flywheel is assembled.



⚠ CAUTION ⚠

The flywheel mounting capscrews must be a minimum of SAE Grade 8 with rolled threads. Use identical replacements to avoid possible flywheel failure resulting in personal injury or property damage.



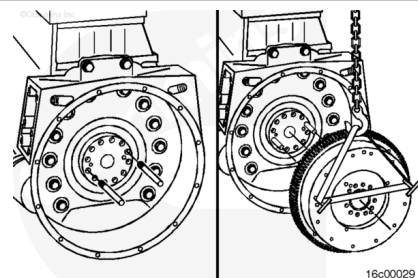
Lubricate the threads and underneath the heads of the capscrews with clean 15W-40 engine oil.

Allow the excess oil to drip off the parts.

Do **not** lubricate the threads of the crankshaft.

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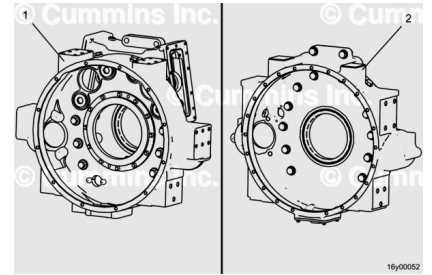
Use two M16 x 1.5 guide studs during assembly.

Install the flywheel.

The flywheel **must** be firmly against the crankshaft.

This engine family has historically been built with two different flywheel mounting options, with a REPTO flywheel housing (1) and a standard flywheel housing (2).

Use the flywheel housing to identify which is used.



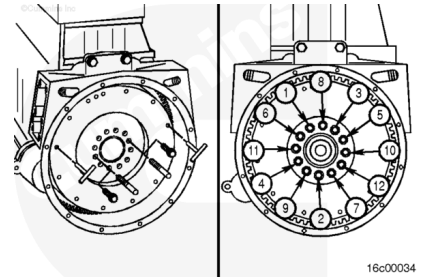
Install the ten capscrews.

Remove the t-handles and the guide studs.

Install the remaining two capscrews into the holes from which the guide studs were removed.

Use flywheel capscrew socket, Part Number 2892191, to tighten the capscrews in the sequence illustrated.

Install and torque all capscrews according to the sequence shown. Use torque value specific to the flywheel mounting option used.



Torque Value:

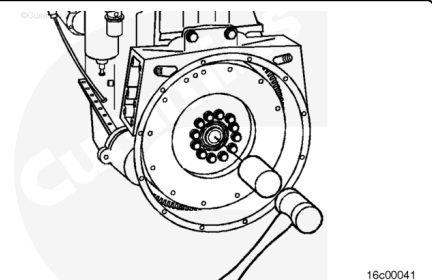
Standard Flywheel Housing 251 n•m [185 ft-lb]

Torque Value:

REPTO Flywheel Housing 298 n•m [220 ft-lb]

Use a mandrel and hammer to install the pilot bearing.

The pilot bearing **must** be installed even with the pilot bore surface.



Measure

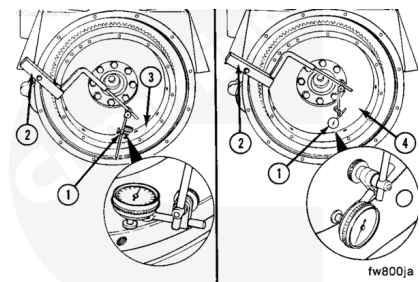
Bore Runout

Use dial indicator gauge (1), Part Number 3376050, or equivalent, and dial gauge attachment (2), Part Number ST-1325, to inspect the flywheel bore (3) and face (4) runout.

Install the attachment to the flywheel housing.

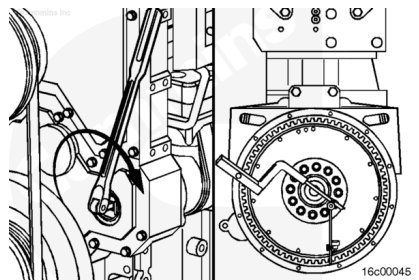
Install the gauge on the attachment.

Install the contact tip of the indicator against the inside of the flywheel bore.



Use the compressor drive or barring device to rotate the crankshaft one complete revolution (360 degrees).

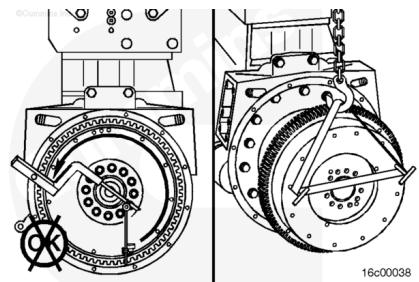
Measure the bore runout.



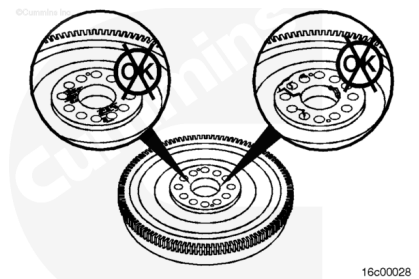
Flywheel Bore Runout

mm		in
0.127	MAX	0.005

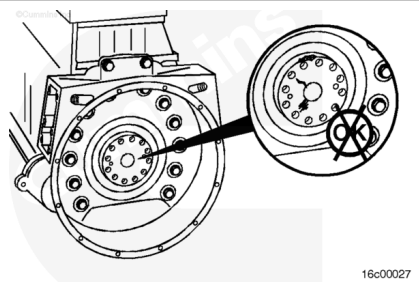
If the total indicator reading is greater than the specification, remove the flywheel.



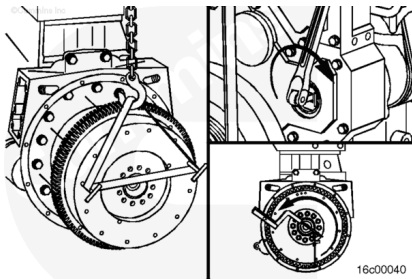
Inspect the flywheel mounting surface for dirt or damage.



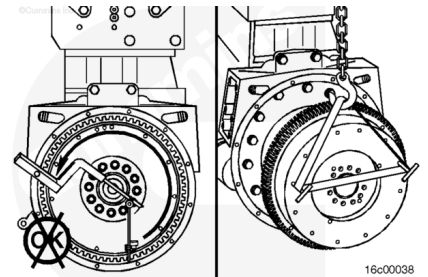
Inspect the crankshaft for dirt or damage.



Install the flywheel and measure the bore runout again.



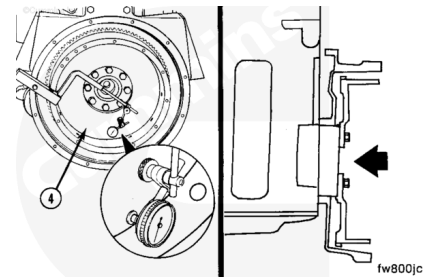
Replace the flywheel if the runout does **not** meet specifications.



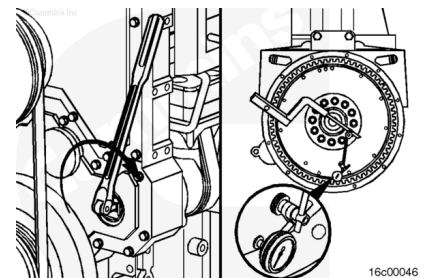
Face Runout

Install the contact tip of the indicator against the flywheel face as close to the outside as possible to inspect the face (4) runout.

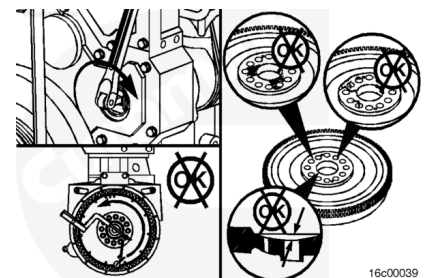
Push the flywheel forward to remove the crankshaft end thrust.



Use the compressor drive or barring device to rotate the crankshaft one complete revolution (360-degrees) while keeping the crankshaft end thrust pushed forward.



The total indicator reading **must not** exceed the following specifications:



Flywheel Face Runout Total Indicator Reading (TIR)	
Flywheel Radius (A)	Maximum TIR
203 mm [8 in]	0.203 mm [0.008 in]
254 mm [10 in]	0.254 mm [0.010 in]
305 mm [12 in]	0.305 mm [0.012 in]
356 mm [14 in]	0.356 mm [0.014 in]

Flywheel Face Runout Total Indicator Reading (TIR)	
Flywheel Radius (A)	Maximum TIR
406 mm [16 in]	0.406 mm [0.016 in]

Finishing Steps

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- Install the clutch discs, pressure plate, transmission, and driveline, if equipped, in reverse order of removal. See equipment manufacturer service information.
- Align the universal joints on each end of the driveshaft to prevent vibration. See equipment manufacturer service information.
- Connect batteries and air starter. See equipment manufacturer service information.
- Start the engine. Check the operation of the equipment.